Actuation-Aware Sensor Validation for Fault-Resilient Closed-Loop Hydroponic Control

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Abstract

Closed-loop hydroponic systems can convert transient sensor faults into unnecessary dosing, cooling, pumping, or carbon dioxide actuation. This paper presents AASVR, an actuation-aware sensor validation and rectification layer for fault-resilient hydroponic control. AASVR combines robust range/rate gates, uncommanded-trend and stuck-value checks, persistence states, bounded rectification, and actuation authorization. On held-out hydroponic replay with synthetic faults, AASVR achieved 0.824 balanced accuracy and 0.259 false actuations per trial, nearly matching the strongest detector while reducing false actuation by an order of magnitude. The results support actuation-aware validation as a practical reliability layer for low-cost closed-loop automation.

Keywords: sensor validation, data rectification, fault detection, process control, hydroponics, actuation

1. Introduction

Hydroponic controlled-environment systems depend on continuous measurement and actuation. In a recirculating Nutrient Film Technique (NFT) installation, the same data stream that describes pH, electrical conductivity (EC), air temperature, humidity, water temperature, water level, and carbon dioxide (CO2) can also trigger dosing pumps, chillers, valves, cooling

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devices, and circulation equipment. A faulty measurement is therefore not only a data-quality issue. It can become a physical command that perturbs the nutrient solution or growth environment.

This coupling is especially important in low-cost hydroponic deployments because sensors may spike, drift, saturate, drop out, or respond slowly after a real actuator command. A single high-pH spike should not immediately authorize acid dosing; a temporary EC excursion should not be treated the same as a persistent concentration shift; and water-level values without defensible calibration should not be allowed to dominate the control analysis. These examples motivate a validation layer that is aware of the downstream actuation decision.

Conventional filtering methods are useful, but they usually ask whether a measurement looks noisy or anomalous in the signal domain. They do not directly ask whether the value is trustworthy enough to update the control state or authorize an actuator. We therefore introduce *actuation-aware sensor validation and rectification* as the central problem for fault-resilient hydroponic control. The key observation is that a value may be worth logging for diagnosis while still being too untrusted to drive a pump, valve, or cooler.

We present a closed-loop NFT hydroponic system as the primary engineered platform and propose AASVR as its central fault-resilience method. AASVR combines robust plausibility gates, persistence states, bounded rectification, and explicit actuation authorization. Public HAI and SKAB process datasets are retained as supplementary stress tests for the same validation interface, but they are not used to support the main hydroponic claim.

The contributions are as follows. First, we describe an integrated closed-loop hydroponic sensing and control platform with separate growth and nutrient-solution modules. Second, we formulate sensor-fault handling in that platform as an actuation-aware validation and rectification problem. Third, we implement AASVR as a streaming state-machine method with bounded updates and explicit actuation authorization. Fourth, we evaluate the method on real hydroponic time series with held-out synthetic faults, ablations, and paired statistical comparisons. Fifth, we provide a reproducible Docker and LaTeX workflow with dataset descriptors and paper-generation artifacts.

2. Related Work

2.1. Sensor faults in closed-loop systems

Sensor data-quality reviews distinguish missingness, outliers, drift, stuck values, saturation, and inconsistent rates as recurring faults in deployed systems [1]. Agricultural IoT systems face the same problem when sensor failure or miscalibration propagates into monitoring or control decisions [2, 3]. In closed-loop hydroponics, these faults have downstream consequences because an erroneous measurement can change the actuator command. This motivates treating validation, rectification, and actuation as separate decisions rather than as one smoothing step.

2.2. Filtering and monitoring baselines

The baseline family used in this paper covers common online filters, process-monitoring methods, model-based statistical tests, and lightweight one-class anomaly detectors. Hampel-style robust outlier detection uses robust scale estimates to suppress isolated deviations [4]. Kalman filtering gives a probabilistic recursive estimator for linear dynamical systems [5]. EWMA and CUSUM-style monitoring are standard approaches for persistent shifts [6, 7]. Generalized likelihood ratio (GLR) tests provide a classical statistical framework for abrupt change detection [8]. PCA and recursive PCA-style residual monitoring use residual statistics to capture coordinated changes or adaptive operating regimes [9, 10, 11]. Isolation Forest, One-Class SVM, and local outlier factor add common one-class machine-learning comparisons for anomaly detection [12, 13, 14]. AASVR is complementary to these methods because it adds a control-facing authorization decision.

2.3. External process datasets

Public process datasets provide independent stress tests for measurement validation. HAI 21.03 contains industrial-control attack data from a hardware-in-the-loop testbed [15]. SKAB contains multivariate process-loop experiments with anomaly and changepoint labels [16]. SWaT, WaDi, and Tennessee Eastman are relevant larger benchmarks for future validation, subject to access and redistribution constraints [17, 18, 19].

3. Hydroponic System Design and Deployment

The target platform is a closed NFT hydroponic system for lettuce growth. NFT systems recirculate a shallow nutrient stream through sloped growth channels, allowing water and nutrient reuse when solution management is reliable [20, 21]. The deployed system is organized into two physical modules. The Growth Module houses the plants, lighting, air-temperature and humidity sensing, CO2 sensing, visual monitoring, and environmental actuation. The Nutrient Solution Container Module stores and recirculates the nutrient solution and contains the water-quality sensors and nutrient-solution actuators.

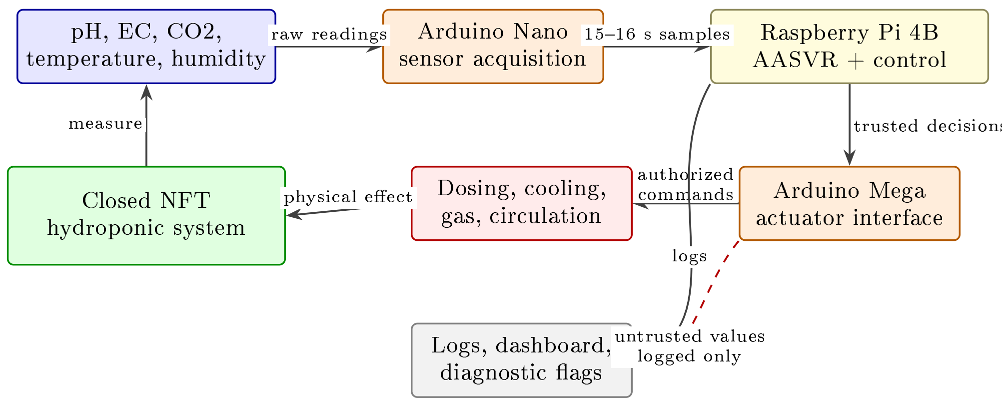


Figure 1: Hydroponic system architecture used as the primary deployment platform. Sensors in the Growth Module and Nutrient Solution Container Module are acquired through the Arduino Nano and sent to the Raspberry Pi 4B at approximately 15-second cadence. The Raspberry Pi logs data, updates the dashboard, runs the validation/control logic, and sends only authorized commands to the Arduino Mega actuator interface. AASVR is positioned between raw sensing and actuation so that untrusted values can be logged without directly triggering pumps, valves, chillers, or cooling devices.

The Growth Module uses a DHT22 sensor for air temperature and humidity and an MH-Z19C nondispersive infrared CO2 sensor for gas concentration. CO2 regulation is implemented through a normally closed pneumatic solenoid valve connected to a CO2 cylinder. Air-temperature regulation is supported by a thermoelectric cooler. Two 40 W LED grow-light arrays provide a controlled light cycle, and a camera provides visual monitoring of the crop during the growth cycle.

The Nutrient Solution Container Module uses an Atlas Scientific analog pH kit, an Atlas Scientific Mini Conductivity K 1.0 kit for EC, a water-

Figure 2: Physical hydroponic system during operation. The photograph is included to show that the evaluation data come from a deployed closed-loop apparatus rather than from a purely simulated process.

temperature sensor, and a JSN-SR04T waterproof ultrasonic distance sensor for water level. pH correction is implemented through acid/base dosing, and EC correction through concentrated nutrient solution or distilled-water addition. After a pH or EC dosing action, the dosing subsystem is disabled for a 10-minute mixing interval before further dosing is allowed. A refrigerative water chiller regulates nutrient-solution temperature, an aerator runs throughout the growth cycle to support dissolved oxygen, and a water pump recirculates nutrient solution through the NFT channels.

The control unit connects sensors to an Arduino Nano and actuators to an Arduino Mega. Both microcontrollers communicate with a Raspberry Pi 4 Model B, which serves as the central control computer. This sensor-microcontroller-edge-computer structure is consistent with prior hydroponic automation systems, but the present paper focuses on the measurement-to-actuation reliability layer rather than on dashboarding as a contribution [22, 23]. The Raspberry Pi receives median-filtered sensor readings from the Arduino Nano, logs the readings, updates the ThingSpeak-backed dashboard, applies the validation/control logic, and sends authorized actuation commands to the Arduino Mega. The dashboard and camera feed are retained as deployment infrastructure; they are not claimed as the research contribution of this paper.

Table 1: Hydroponic platform components relevant to measurement validation and actuation. Water level is listed because it is part of the deployed system, but it is excluded from the primary quantitative analysis because the available raw values are not defensibly calibrated.

Subsystem	Measurements	Actuation or use
Growth module	Air temperature, humidity	Thermoelectric cooling
Growth module	CO2 concentration	CO2 solenoid valve
Growth module	Camera feed	Visual crop monitoring
Growth module	Light schedule	LED grow lights
Solution module	pH	Acid/base dosing, 10 min lockout
Solution module	EC	Nutrient/water dosing, 10 min lockout
Solution module	Water temperature	Refrigerative chiller
Solution module	Water level	Warning only in this analysis
Solution module	Recirculation state	Pump and aerator operation

The evaluation below treats this system as a deployed measurement-to-actuation stream: raw measurements are replayed through validation methods, transformed into trusted states, and then passed through an authorization rule before any actuator command is counted.

4. Problem Formulation

For hydroponic sensor s at time t, let $y_s(t)$ be the raw measurement and $\hat{x}_s(t)$ be the trusted value exposed to control. The method observes recent history $Y_s(t-w:t)$, broad physical limits $[P_{s,\min}, P_{s,\max}]$, a control band $[L_s^a, U_s^a]$, a rate limit r_s, and any available actuator state $a(t)$. In this system, s may correspond to pH, EC, humidity, air temperature, water temperature, or CO2, while $a(t)$ may represent dosing, cooling, circulation, or gas-release activity.

Table 2: Notation used in the AASVR formulation. Variables are defined for one sensor stream and then applied independently or jointly across the hydroponic sensor set.

Symbol	Meaning
$y_s(t)$	Raw value from sensor s at sample t
$\hat{x}_s(t)$	Trusted value exposed to the controller
$Y_s(t-w:t)$	Rolling window of recent raw values
$[P_{s,\min}, P_{s,\max}]$	Broad physically plausible range
$[L_s^a, U_s^a]$	Actuation or control band
r_s	Maximum plausible rate of change
u_s	Sensor uncertainty or minimum measurement resolution term
$q_s(t)$	Trust score in $[0, 1]$
$z_s(t)$	Sensor state in the AASVR state set
$h_s(t)$	Evidence from the recent persistence/confirmation window
$A_s(t)$	Binary actuation authorization flag
θ	Tunable AASVR parameter vector

We define a measurement as *untrusted* when it is missing, physically implausible, rate-inconsistent, inconsistent with recent trusted state, or inconsistent with expected actuator effects. We define a *false actuation* as an authorization caused by untrusted evidence. A *missed actuation* is a failure to authorize action when the trusted state remains outside the control band.

The validation decision is separated from the control decision. Let $g_s(t) \in \{0, 1\}$ denote whether a raw sample passes all measurement gates, and let $z_s(t)$ denote one of five states: normal, suspect transient, persistent deviation, accepted regime shift, and fault alert. The controller receives $\hat{x}_s(t)$ and $A_s(t)$, not the raw value alone. This separation is necessary because a value can be physically possible but still too uncertain to authorize actuation.

The evaluation follows a control-aware objective,

$$
\begin{aligned}
J(\theta) = & w_1 E_{\text{pass}} + w_2 E_{\text{reject}} + w_3 A_{\text{false}} + w_4 A_{\text{missed}} \\
& + w_5 D_{\text{delay}} + w_6 T_{\text{unsafe}} + w_7 C_{\text{compute}},
\end{aligned} \tag{1}
$$

where the terms respectively measure anomalous pass-through, false rejection, false actuation, missed actuation, detection delay, residence outside safe bands, and computational cost. In the implemented validation sweep, the reported scalar objective uses normalized benchmark outputs with weights $w_{\text{BA}} = 1.0$, $w_{\text{false act}} = -0.01$, $w_{\text{false alarm}} = -0.0005$, and $w_{\text{delay}} = -0.001$; the other terms are logged but not used for parameter selection in the current experiments.

5. Proposed Method: AASVR

5.1. Overview

Figure 3 summarizes the AASVR pipeline. The method separates four decisions that are often conflated: measurement validation, trusted-state update, rectification, and actuation authorization. This separation is the core design choice. In the hydroponic system, this means that a suspicious pH value can be stored with a diagnostic flag while acid/base dosing remains suppressed until the value is persistent and trusted.

5.2. Robust plausibility gate

For each sensor, AASVR estimates a robust scale over adjacent changes,

$$
\sigma_{\Delta,s}(t) = 1.4826\,\text{MAD}\left(\Delta y_s(t-w:t)\right). \tag{2}
$$

The accepted deviation tolerance is

$$
\xi_s(t) = \max(\xi_{s,\min}, c_s \sigma_{\Delta,s}(t), r_s \Delta t, u_s). \tag{3}
$$

A reading must satisfy broad physical range, maximum rate, and trusted-state deviation gates. The three elementary gates are

$$
\begin{aligned}
G_s^P(t) &= \mathbf{1}\{P_{s,\min} \le y_s(t) \le P_{s,\max}\}, \\
G_s^R(t) &= \mathbf{1}\{|y_s(t) - y_s(t-1)|/\Delta t \le r_s\}, \\
G_s^D(t) &= \mathbf{1}\{|y_s(t) - \hat{x}_s(t-1)| \le \xi_s(t)\}.
\end{aligned} \tag{4}
$$

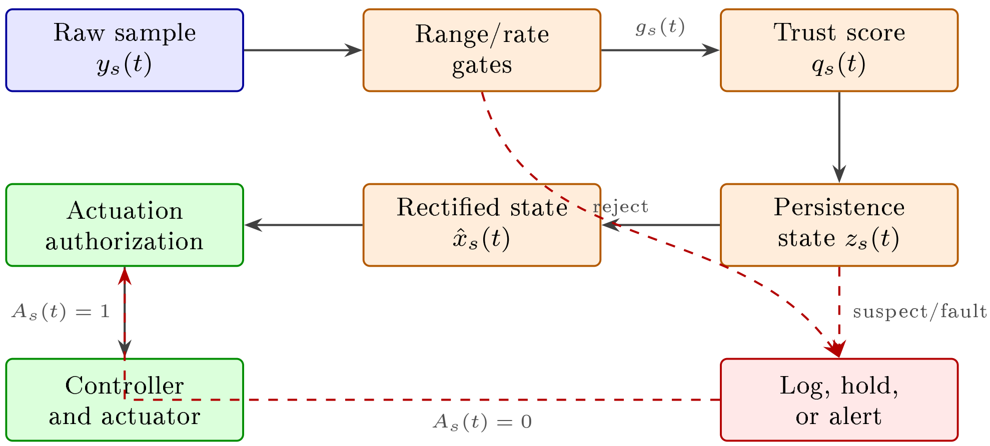


Figure 3: Architecture of AASVR. Raw measurements first pass through robust plausibility gates that check physical range, local rate, trusted-state deviation, and uncommanded trends. Accepted or suspect readings then update a persistence state machine, which decides whether the trusted state should be updated, held, or rectified by a bounded prediction. Only trusted out-of-band states can authorize actuation. Rejected or fault-alert readings are logged and may trigger maintenance alerts, but they do not directly drive the actuator.

A separate trend guard $G_s^T(t)$ rejects slow monotonic movement when no relevant actuator activity explains the direction of change. A stuck-value guard $G_s^S(t)$ rejects sustained exact repetition after a preceding change, which targets frozen plausible readings without treating ordinary low-variance noise as a fault. The overall gate is therefore

$$g_s(t) = G_s^P(t)G_s^R(t)G_s^D(t)G_s^T(t)G_s^S(t)G_s^M(t), \tag{5}$$

where $G_s^M(t)$ is one only when the value is present and parseable. The broad physical range is deliberately different from the crop-control band: for example, a pH value can be undesirable for lettuce while still being physically possible and therefore not automatically a sensor fault.

5.3. Trust score

AASVR also computes a continuous trust score so that control can distinguish weakly plausible values from strongly confirmed values. The implementation uses a weighted additive score,

$$q_s(t) = \sum_{j \in \mathcal{Q}} w_j q_{j,s}(t), \quad \sum_{j \in \mathcal{Q}} w_j = 1, \tag{6}$$

where $\mathcal{Q}$ contains range, rate, persistence, actuator-consistency, missingness, and trend components. Each component is scaled to $[0, 1]$. In the hydroponic controller, $q_s(t)$ is used together with persistence and cooldown logic rather than as a standalone anomaly score.

5.4. State-machine persistence logic

Figure 4 shows the five persistence states. A single failed gate moves a sensor into a suspect transient state. Repeated failures create a persistent-deviation state. Consistent evidence can be accepted as a new process regime, while missing, stuck, saturated, or contradictory readings escalate to fault alert. This matters for hydroponics because real mixing and thermal responses can take multiple samples after dosing or cooling, whereas isolated spikes should not immediately change the trusted state.

The state update can be written as

$$z_s(t) = \mathcal{T}\left(z_s(t-1), g_s(t), q_s(t), h_s(t), a(t)\right), \tag{7}$$

where $h_s(t)$ summarizes persistence-window evidence. This representation makes the state machine explicit without treating every transition as a separate hand-tuned rule in the manuscript. The transition function $\mathcal{T}$ is implemented by Algorithm 1.

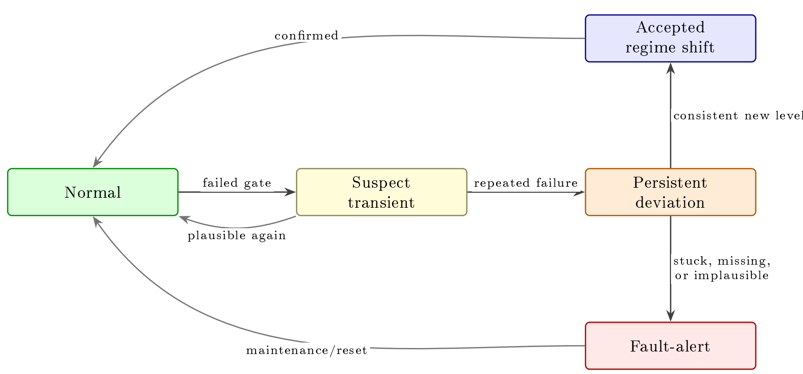


Figure 4: AASVR state machine. The state machine prevents one failed sample from immediately becoming a maintenance alert, but also prevents repeated failed samples from silently entering the controller. The accepted-regime-shift path allows real process changes to be incorporated after confirmation. The fault-alert path blocks single-sensor actuation authorization until maintenance, reset, or independent evidence restores trust.

5.5. Rectification and actuation authorization

When a reading is accepted, AASVR sets $\hat{x}_s(t) = y_s(t)$. When it is suspect, the method either holds $\hat{x}_s(t-1)$ or applies a bounded update limited by $r_s \Delta t$:

$$\hat{x}_s(t) = \begin{cases} y_s(t), & z_s(t) \in \{\text{normal}, \text{accepted}\}, \\ \hat{x}_s(t-1), & z_s(t) = \text{suspect}, \\ \hat{x}_s(t-1) + \text{clip}(\Delta y_s, -r_s \Delta t, r_s \Delta t), & \text{bounded}, \\ \text{alert}, & z_s(t) = \text{fault}. \end{cases} \tag{8}$$

Actuation is authorized only if the trusted value is outside the control band, the violation persists for the configured number of samples, the sensor state is normal or accepted-regime-shift, the trust score is sufficient, and the actuator is not in cooldown:

$$\begin{aligned} A_s(t) = & \mathbf{1}\{\hat{x}_s(t) \notin [L_s^a, U_s^a]\} \mathbf{1}\{p_s(t) \geq k_s\} \mathbf{1}\{q_s(t) \geq q_{\min}\} \\ & \times \mathbf{1}\{z_s(t) \in \mathcal{Z}_{\text{safe}}\} \mathbf{1}\{\ell_s(t) = 0\}. \end{aligned} \tag{9}$$

Here $p_s(t)$ is the number of consecutive trusted control-band violations, k_s is the persistence requirement, $\mathcal{Z}_{\text{safe}} = \{\text{normal}, \text{accepted}\}$, and $\ell_s(t)$ is the actuator lockout indicator. For pH and EC, this rule complements the 10-minute dosing lockout already present in the deployed system; for CO2 and temperature, it prevents transient readings from becoming immediate gas-release or cooling commands.

The bounded branch above also gives a simple boundedness guarantee for the value exposed to control. If the true process state has sample-to-sample change bounded by $r_s \Delta t$ and the previous trusted estimate has finite error, then the bounded rectification update cannot move faster than the same physical-rate envelope. Thus a rejected raw value cannot drive $\hat{x}_s(t)$ outside a rate-limited reachable set in one step. This is not a full closed-loop stability proof, but it prevents estimator divergence caused by a single large measurement excursion and gives the controller a bounded input while the sensor remains suspect.

Algorithm 1 Streaming AASVR update for one sensor

```
Require: Raw value y_s(t), previous trusted state x̂_s(t − 1), recent history, limits, actuator state.
Estimate robust local scale σ_{Δ,s}(t) and tolerance ξ_s(t).
Apply physical-range, rate, trusted-deviation, missingness, trend, and stuck-value gates.
Update the persistence state from the gate outcome.
if the reading is accepted then
    Set x̂_s(t) ← y_s(t).
else if the reading is transient suspect then
    Hold or bounded-predict x̂_s(t).
else
    Keep the last safe estimate and log a fault-alert decision.
end if
Authorize actuation only if trusted state, persistence, score, and cooldown rules are satisfied.
return Trusted value, sensor state, trust score, actuation flag, and log record.
```

6. Evaluation Protocol

6.1. Hydroponic deployment data

The hydroponic data contain 121244 raw rows collected from 2024-02-27 to 2024-03-28 at a median cadence of approximately 16 seconds. The primary analysis uses EC, pH, humidity, air temperature, water temperature, and CO2. Water level is part of the deployed system but is excluded from the primary analysis because the available raw values are not defensibly calibrated.

6.2. Supplementary external stress tests

HAI 21.03 and SKAB are retained as supplementary stress tests rather than as main evidence. Their labels describe cyber-physical attacks and process-loop anomalies, not hydroponic sensor faults. For these datasets, a timestamp is predicted abnormal when the fraction of rejecting sensor streams exceeds an aggregation threshold selected on the first half of each native-label series and reported on the second half. The results are used only to characterize limits of direct transfer.

6.3. Fault injection protocol

Synthetic faults are injected into windows sampled from the real hydroponic background. For each sensor, fault type, magnitude, and duration, three deterministic replicates are generated and assigned to tune, validation, and test splits. The reported hydroponic benchmark uses only the held-out test split; AASVR and baseline hyperparameters are selected on validation trials. Fault windows include 90 pre-fault samples and 120 post-fault samples so that detection delay and false alarms are evaluated around the injected event rather than in isolation.

Table 3: Synthetic fault-injection protocol for the hydroponic replay benchmark. For each protocol row, durations are 1, 5, and 30 samples with one replicate assigned to each split. Magnitudes are in sensor units: pH uses 0.5/2.0, CO2 500/2000 ppm, EC 100/400, air temperature 2/8 °C, and water temperature 1/4 °C.

Fault type	Injection rule over fault window $\mathcal{I}$	Purpose
Spike	$y'_s(t) = y_s(t) + m$	Single-window offset
Multi-spike	$y'_s(t) = y_s(t) + b_t m,\ b_t \in \{-1, 1\}$	Alternating impulse faults
Dropout	$y'_s(t) = \mathrm{NaN}$	Missing samples
Drift	$y'_s(t) = y_s(t) + m(t - t_0)/\lvert\mathcal{I}\rvert$	Slow monotonic deviation
Bias	$y'_s(t) = y_s(t) + m$	Persistent offset
Noise burst	$y'_s(t) = y_s(t) + \epsilon_t,\ \epsilon_t \sim \mathcal{N}(0, m^2)$	Increased variance
Step	$y'_s(t) = y_s(t) + m$	Abrupt sustained change
Stuck-at	$y'_s(t) = y_s(t_0)$	Constant repeated value
Saturation	$y'_s(t) = m$	Sensor rail or implausible fixed value

6.4. Baselines and metrics

The baselines are raw thresholding, moving average, moving median, Hampel, Kalman, EWMA, CUSUM, PCA-style monitoring, GLR, recursive PCA, Isolation Forest, One-Class SVM, and local outlier factor. The earlier manuscript rule is retained in the reproducibility outputs but omitted from the main comparison because it duplicates the raw physical-threshold decision in the current replay protocol. Every displayed baseline is tuned on the validation split using the same control-aware objective family as AASVR and then wrapped in the same persistence/cooldown authorization supervisor for false-actuation scoring. This separates detector quality from downstream authorization. The hydroponic synthetic-fault benchmark reports precision, recall, specificity, balanced accuracy, detection delay, false alarm events, false actuation mean, and false actuation standard deviation.

Bootstrap confidence intervals use 1000 resamples of injected-fault trials, not individual samples. Paired Wilcoxon signed-rank tests compare AASVR against each baseline on per-trial balanced accuracy and false actuation, with Holm correction for the primary method comparisons. Sensor–fault grids use Benjamini–Hochberg correction.

Table 4: Baseline taxonomy used in the hydroponic benchmark. The final column marks whether a method explicitly separates anomaly detection from actuation authorization.

Family	Methods	Primary role	Act.-aware
Rule	Raw threshold	Existing physical-limit control logic	No
Filters	Moving average, moving median, Hampel, Kalman	Noise and spike suppression	No
Process monitoring	EWMA, CUSUM, PCA, GLR, recursive PCA	Shift or process-deviation detection	No
One-class ML	Isolation Forest, One-Class SVM, LOF	Data-driven anomaly boundary	No
Proposed method	AASVR	Validation, rectification, and authorization	Yes

7. Results

7.1. Hydroponic deployment characterization

The deployment data are first used to characterize the real measurement background on which AASVR operates. The 121244-row feed covers a continuous 30-day lettuce growth period, with approximately 16-second median sampling. The key point for this paper is not that the dashboard collected data, but that the closed-loop system generated the kind of sensor-to-actuator stream in which a bad reading can trigger a physical correction.

7.2. Hydroponic synthetic-fault benchmark

Table 6 summarizes the primary target-domain result on held-out synthetic-fault trials. LOF and AASVR are effectively tied on balanced accuracy, but they reach that point through different tradeoffs: LOF has higher recall and lower specificity, whereas AASVR has much lower false actuation under the shared supervisor. This is the intended comparison after

Table 5: Hydroponic measurement streams used in the primary AASVR analysis. The table separates deployed sensing from quantitative use so that the water-level exclusion is explicit rather than hidden.

Variable	Module	Primary analysis	Control relevance
EC	Solution	Yes	Nutrient/water dosing evidence
pH	Solution	Yes	Acid/base dosing evidence
Humidity	Growth	Yes	Growth-environment monitoring
Air temperature	Growth	Yes	Air-cooling evidence
Water temperature	Solution	Yes	Chiller evidence
CO2	Growth	Yes	CO2 valve evidence
Water level	Solution	No	Warning channel; calibration unresolved

the reviewer-suggested fairness correction: the detector component is competitive, while the authorization layer produces the operational reduction in unnecessary actuation.

Table 6: Held-out hydroponic synthetic-fault benchmark. Baselines are validation-tuned and evaluated on the same test split with a shared persistence/cooldown supervisor. Bold entries mark the best value in each metric column among the displayed methods; lower is better for false actuation.

Method	Recall	Specificity	Bal. acc.	FPR	Delay	False act.	False act. SD
LOF	0.853	0.798	**0.826**	0.202	0.270	2.811	4.240
AASVR	0.777	0.871	0.824	0.129	**0.000**	**0.259**	**0.853**
Kalman	0.613	0.803	0.708	0.197	**0.000**	0.515	1.175
Isolation Forest	0.581	0.826	0.703	0.174	0.126	2.907	4.253
Hampel	0.411	0.984	0.698	0.016	**0.000**	1.041	1.502
Moving median	0.411	0.984	0.698	0.016	**0.000**	0.993	1.496
Recursive PCA	0.317	0.988	0.652	0.012	**0.000**	1.319	1.653
One-Class SVM	**0.912**	0.333	0.623	0.667	0.089	2.796	4.227
GLR	0.129	**0.999**	0.564	**0.001**	**0.000**	1.330	1.638

Table 7: Paired held-out effects for AASVR against representative baselines. Positive ΔBA means AASVR has higher per-trial balanced accuracy; positive Δfalse actuation means the baseline has more false actuations than AASVR. P-values are Holm-corrected paired Wilcoxon tests.

Baseline	ΔBA	p_{BA}	ΔFalse act.	95% CI	p_{FA}
LOF	-0.002	0.0025	2.552	[2.077, 3.026]	6.23×10^{-22}
Isolation Forest	0.121	1.85×10^{-13}	2.648	[2.192, 3.107]	6.66×10^{-23}
Kalman	0.116	4.34×10^{-19}	0.256	[0.107, 0.393]	5.92×10^{-4}
Hampel	0.126	1.51×10^{-8}	0.781	[0.630, 0.937]	3.31×10^{-15}
CUSUM	0.184	8.19×10^{-19}	1.015	[0.844, 1.185]	2.34×10^{-20}

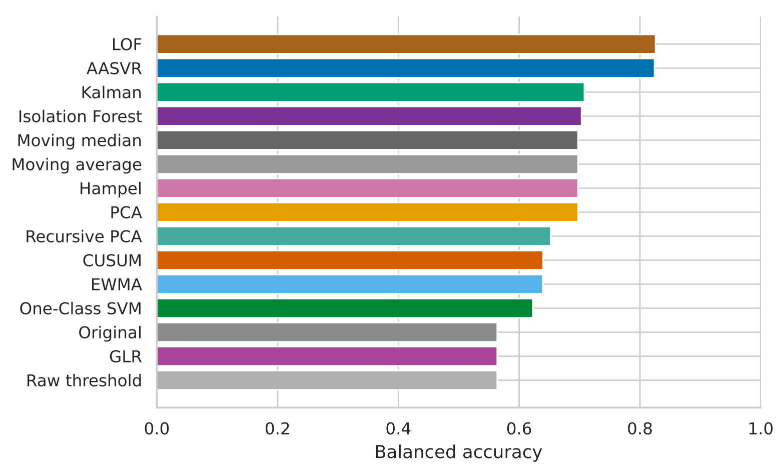


Figure 5: Held-out hydroponic synthetic-fault balanced accuracy across methods. The comparison is computed on test split injected faults over windows sampled from the real hydroponic data after validation-only hyperparameter tuning. AASVR is competitive with LOF on detector score while providing substantially lower false actuation.

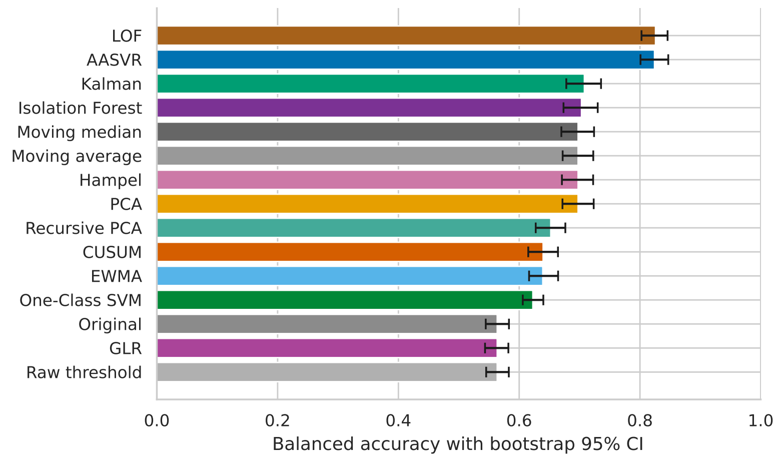


Figure 6: Bootstrap uncertainty for held-out hydroponic synthetic-fault balanced accuracy. Intervals are computed by resampling injected-fault trials. The result should be read together with the paired tests because AASVR and LOF are close on point-wise balanced accuracy, while false-actuation behavior differs strongly.

7.3. Sensor-wise and fault-type comparisons

Table 8 breaks down AASVR performance by hydroponic sensor. CO2, water temperature, and air temperature faults are detected with balanced accuracy above 0.84. EC is the weakest stream, mainly because gradual concentration faults are harder to distinguish from plausible nutrient-solution changes without an independent reference.

Table 9 and Figure 7 show performance by injected fault class. AASVR performs best on saturation, dropout, bias, spike, noise burst, step, and multi-spike faults. Drift and stuck-at faults remain difficult because they can resemble slow process movement or repeated steady values without an independent reference. The added stuck-value guard improves stuck-at balanced accuracy only modestly, from 0.465 without the guard to 0.474 with the guard, so it should be read as a mitigation rather than a solved failure mode.

Table 8: AASVR synthetic-fault performance by hydroponic sensor. False actuation is reported as mean events per injected-fault trial.

Sensor	Recall	Specificity	Bal. acc.	False act.
CO2	**0.849**	**0.884**	**0.867**	0.259
Water temperature	0.820	0.873	0.847	0.222
Air temperature	0.822	0.863	0.842	0.222
pH	0.785	0.870	0.828	**0.019**
EC	0.609	0.865	0.737	0.574

Table 9: AASVR synthetic-fault performance by injected fault type. Stuck-at and drift are the most difficult conditions because frozen plausible values and slow monotonic movement can resemble real process behavior.

Fault type	Recall	Specificity	Bal. acc.	False act.
Saturation	**1.000**	0.866	**0.933**	**0.000**
Dropout	**1.000**	0.859	0.930	**0.000**
Bias	0.910	0.871	0.890	0.233
Spike	0.896	**0.880**	0.888	0.167
Noise burst	0.893	0.878	0.886	0.133
Step	0.876	0.879	0.877	0.233
Multi-spike	0.879	0.871	0.875	**0.000**
Drift	0.468	0.861	0.665	0.233
Stuck-at	0.072	0.875	0.474	1.333

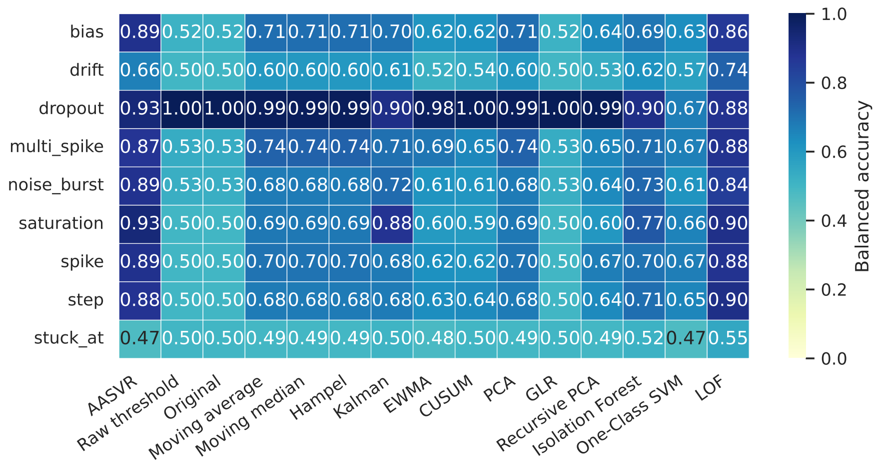


Figure 7: Hydroponic synthetic-fault heatmap by sensor and fault type. The figure summarizes where AASVR is strongest and where slow drift remains a limitation.

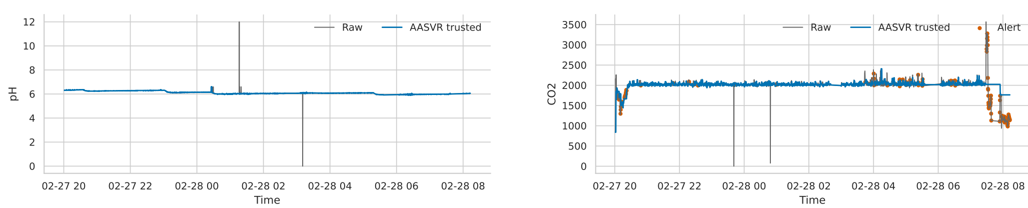
7.4. Ablation and control effect

Table 10 shows that several ablations preserve similar detection scores but change control behavior. Removing actuation confirmation increases false actuation from 0.233 to 0.583 events per trial in the sampled held-out ablation set. The ablation uses a deterministic 120-trial subset of the held-out test split for tractability, which explains the small difference from the full-test result in Table 6. The robust-MAD and stuck-detector variants differ only slightly from the full model on this subset, so those rows should be interpreted as sensitivity checks rather than evidence of a large component effect.

The validation sweep in Table 11 shows that the selected AASVR configuration is not an isolated optimum. Multiple trust-score and transient-limit settings produced the same validation objective when the robust scale multiplier was 5.0. These parameters were selected before evaluating the held-out test split in Table 6.

7.5. Representative hydroponic trace

Figure 8 shows replay segments from the hydroponic deployment. The traces are included to connect the aggregate benchmark to the actual deployment setting. They illustrate the measurement-to-state transformation that precedes any actuation decision.



(a) pH replay (b) CO2 replay

Figure 8: Representative hydroponic replay traces. The raw streams and AASVR trusted estimates are plotted over early deployment segments. These traces are not used as ground-truth fault labels; they show how the method exposes a trusted state rather than passing every raw measurement directly to the controller.

Table 10: AASVR ablation summary on sampled held-out test trials. Detection scores can remain similar while control outcomes change. In particular, removing actuation confirmation leaves balanced accuracy essentially unchanged but increases false actuation, which supports evaluating control-aware metrics rather than point-wise detection alone.

Variant	Bal. acc.	Recall	False act.
Full AASVR	**0.820**	0.765	0.233
No stuck detector	0.819	0.757	0.233
No robust MAD scale	0.819	**0.794**	**0.192**
No actuation confirmation	**0.820**	0.765	0.583
No cooldown	**0.820**	0.765	0.475
No trend guard	0.804	0.728	0.325
Bounded prediction mode	0.774	0.669	0.575

Table 11: Top AASVR parameter settings from the validation split. The selected setting uses $c_s = 5.0$, $q_{\min} = 0.70$, and transient limit 2; tied settings are retained in the table to show sensitivity.

c_s	$q_{\min}$	Transient limit	Bal. acc.	Objective
5.0	0.85	3	**0.827**	**0.750**
5.0	0.85	2	**0.827**	**0.750**
5.0	0.70	3	**0.827**	**0.750**
5.0	0.70	2	**0.827**	**0.750**
4.0	0.85	3	0.826	0.749

7.6. Agronomic operation evidence

The lettuce growth trials are retained as deployment evidence, not as causal proof that AASVR improved crop yield. The automated treatment differed from the soil and manual DWC controls in enclosure, lighting, cooling, nutrient delivery, CO2 regulation, and automation. The agronomic data therefore show that the closed-loop apparatus operated through complete growth cycles while supporting the method evaluation; detailed plant-growth plots are moved to supplementary material.

8. Discussion

8.1. Why actuation awareness matters

The ablation and baseline results show why actuation-aware validation should be evaluated with control metrics. The detection component is competitive with tuned baselines on held-out synthetic faults, but the authorization layer is what produces the operationally important false-actuation reduction. For hydroponic systems, this distinction is practical because unnecessary acid/base dosing, nutrient dosing, CO2 release, or cooling can disturb the process even if the signal-level anomaly score appears acceptable.

8.2. Cross-domain generalization

Supplementary HAI and SKAB stress tests show that AASVR can run on independent industrial-control and process-loop datasets with the same streaming interface, but direct transfer is weak even after validation-selected aggregation thresholds. HAI attacks and SKAB process anomalies are not identical to low-cost hydroponic sensor faults. These results therefore reveal limits of direct parameter transfer and support treating AASVR primarily as a hydroponic measurement-to-actuation reliability layer unless process-specific external models are added.

8.3. Limitations

The hydroponic experiment does not provide independent reference instruments for every sensor at every time point. Synthetic fault injection is therefore necessary for controlled scoring, but injected faults cannot cover every real failure mode. Slow drift, stuck-at behavior, and EC-bias cases remain the principal weaknesses because they can resemble real nutrient-solution dynamics without cross-sensor or model-based residual evidence. The added stuck-value guard catches only sustained exact freezes after a preceding change; it does not solve plausible frozen values that begin during an already steady process. A stronger deployment version should add actuator-response residuals, for example requiring pH or EC to move in the expected direction after dosing, but the current processed feed does not contain actuator-state columns sufficient to score that extension. Water level is excluded from primary analysis because the available raw values are not defensibly calibrated. Crop outcomes are treated only as deployment context because enclosure, lighting, cooling, nutrient delivery, carbon dioxide regulation, and automation are confounded.

9. Conclusion

We presented a closed-loop hydroponic control system and proposed AASVR as its actuation-aware sensor validation and rectification layer. The method separates validation, state update, rectification, and actuation authorization so that untrusted measurements can be logged without directly driving physical commands. On held-out hydroponic synthetic faults, AASVR was competitive with the strongest tuned detector while producing much lower false actuation. The remaining failures on stuck-at and drift faults show where future work should add actuator-response and cross-sensor residual checks rather than relying only on single-stream plausibility gates.

Data Availability

The repository contains code, configuration files, processed-data descriptors, Docker workflow files, and paper-generation assets. Raw hydroponic and external benchmark files are not committed to the repository. Public datasets can be obtained from their original providers, and request-access datasets should be obtained under the providers' terms.

Declaration of Competing Interest

The authors declare no competing interests for the current manuscript draft.

Declaration of Generative AI and AI-Assisted Technologies

AI-assisted tools were used to help organize code, draft text, and check consistency. The authors remain responsible for the scientific content, claims, results, and final manuscript.

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